

Internal Use Only

High Level Design (HLD)

Data Platform- Federated Model (In discussion with Delivery Leads)

<Project INVP/PRJ>

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Document Guidance

The blue italic text in this document is intended to give guidance and should be removed once the template is completed.

Add sections if needed and put 'Not Applicable' in sections that are not required.

1. Document Control

Document Distribution

Name	Position	Reason for distribution
ARB	Architecture Review Board	Approver (Impact Level 3)
DAA	Technical Design Authority	Approver (Impact Level 2)
SDA	Solutions Design Authority	Approver (Impact Level 1)
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1. Executive summary

DAP is a **shared, governed Databricks data platform** operated by a central team, under **one umbrella** — a single Unity Catalog metastore.

Consumers come in two shapes:

- **In-platform teams** — work inside DAP, get a schema + dedicated compute, read Conform via UC grants.
- **Cross-platform consumers** (e.g. **NESO.AI / Palantir Foundry**) — work on their own AI/ML platform and read DAP's curated data through governed consumption channels.

The operating principle is simple:

DAP performs ALL the heavy / big transformations and publishes curated, governed data (Acquisition → Conform/Silver, with the heavy transformation logic baked in → Reference). Consumers consume that curated data through governed channels and do only lightweight transformations on their own compute. The Conform layer is the contract boundary between the platform and its consumers.

Consumers read DAP data through **four governed consumption modes**, chosen by data size and use case:

- **Serve API** — small data, per-DatasetID, APIM-fronted, SPN-authenticated, authorised via a Custom SQL DB.
- **Direct SQL Warehouse query** — big data / AI-ML, consumer SPN with reader access on the SQL Warehouse.
- **Delta Sharing** — big data, zero-copy, cross-platform (**design in progress**).
- **Power BI** — reporting via an SPN registered against the SQL Warehouse via Shared Data Gateway or DAP Gateway. We are also looking for Dedicated Vnet Data Gateway (Design in Progress)

A consumer's journey is:

1. **Request access** — raise an Entra ID security group in ServiceNow, share it with DAP.
2. **Get provisioned** — DAP creates the team's schema (in-platform) and/or the consumption-mode access (SPN reader on warehouse, Serve API authorisation, Delta share), and stands up sandbox + compute.
3. **Develop in sandbox** — on sample / BYO non-PII data; define the heavy-transform spec for DAP. Consumer team also get access to ADO Board for development.

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4. Promotion Gate — DAP brings the real source in and runs the **heavy transformation** into curated Conform.

5. Switch over — re-point reads from sandbox to Conform tables (a minimal change: adopt DAP's conformed column standards + table paths into your Env (AI platform/Palantir/DAP Prod Workspace)).

6. Consume & light-transform — read curated Conform via a consumption mode, do lightweight transforms on own compute, store the output (own <team> schema in-platform, or own side cross-platform); **optionally re-ingest into DAP (Pattern B)** if it must become governed.

This document specifies the **onboarding workflow, the consumption modes and their authentication, the exact access grants, the data flow, compute, governance, and guardrails** so any consumer can be onboarded the same way, every time.

2. Scope

In scope

- How a consumer gains access to the DAP platform and to governed data.
- The four governed **consumption modes** and their authentication.
- The end-to-end data flow from source to the consumer's output.
- Access provisioning: Entra ID groups, SCIM, Unity Catalog grants, SPNs, Serve API authorisation, Delta Sharing.
- Environments, promotion, compute, governance, guardrails.

Out of scope

- DAP's internal ingestion + heavy-transformation engineering detail (canonical Data Acquisition pattern + per-source HLDs).
- The consumers' own lightweight-transformation logic and on-side storage.
- Deep detail of each consumption mode — see the dedicated HLDs (Serve API, Delta Sharing, Power BI VNet Gateway, ML Studio → SQL Warehouse); this HLD references them.

3. Operating model and principles

#	Principle	Implication
PI	One umbrella	One UC metastore = single source of truth. Read Conform with zero copies.

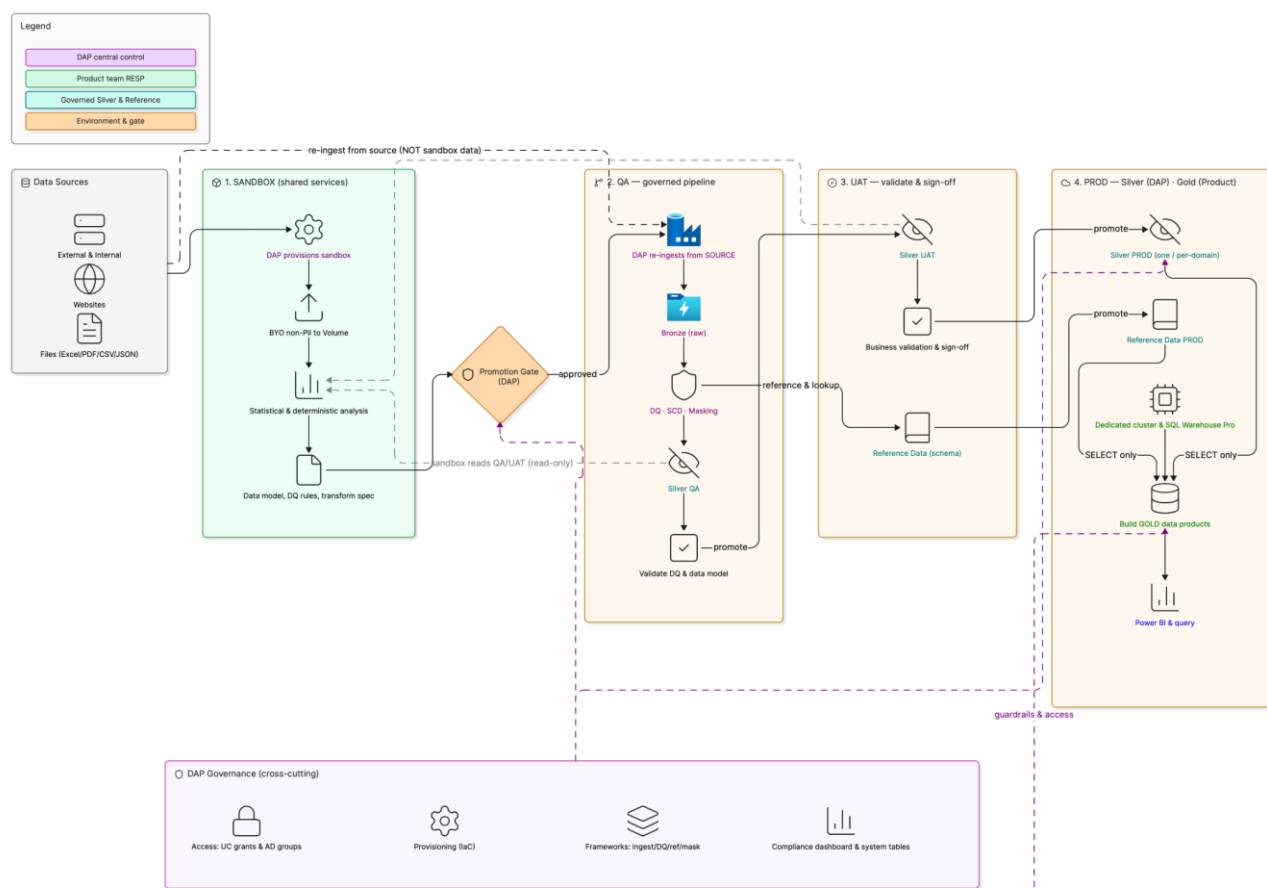
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P2	Schema per team, catalog per environment	<env>_catalog.<team> for in-platform teams. Environment and team both first-class and auditable.
P3	Conform is the Curated boundary	DAP publishes curated Conform tables with stable column standards + paths. Consumers build on them.
P4	DAP does heavy transforms; consumers do light	DAP bakes the heavy / big transformation logic into curated Conform tables. Consumers do only lightweight transforms on their own compute (NESO.AI compute, Palantir compute).
P5	Re-ingest from source, never promote sandbox data	Production data is acquired from the system of record through the governed pipeline (reference data via the loader is the only file-based exception).
P6	Least privilege, scoped grants	SELECT scoped to the specific Conform_Restricted tables a consumer needs (based on Owner Approval) and all Public/Internal Tables— never catalog-wide.
P7	Sandbox is non-PII only	PII enters only through the governed pipeline, masked in Conform.
P8	Self-serve on a paved road	DAP provides frameworks (ingestion, DQ, reference loader, masking), provisioning (IaC), guardrails.

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P9	Consume via governed channels	Four modes — Serve API, direct SQL Warehouse, Delta Sharing, Power BI — chosen by data size + use case; all SPN-authenticated.
P10	Optional feedback loop	A consumer's curated output can be re-ingested into DAP (Pattern B) when it must become governed / shared.

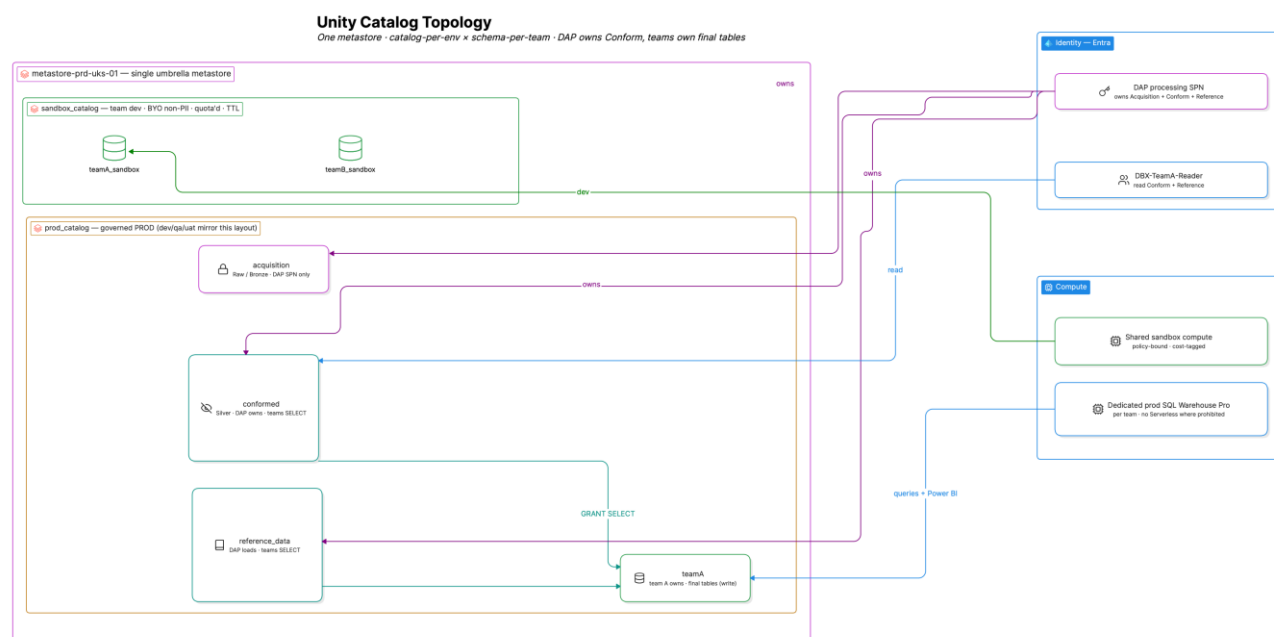
Sample Operating Model RESP team using:



4. Unity Catalog topology

One metastore (metastore-prd-uks-01), **catalog-per-environment × schema-per-team**.

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metastore-prd-uks-01 (the umbrella – single source of truth)

```

|
|—— sandbox_catalog      ← team dev · BYO non-PII · quota'd · TTL · reads QA/UAT (not Prod)
|   └── <team>            (team WRITE; + a UC Volume for BYO files)
|
|—— dev_catalog / qa_catalog / uat_catalog / prod_catalog (one per environment, same layout)
|   ├── acquisition       (Raw / Bronze – DAP SPN ONLY)
|   ├── conformed         (Silver, curated – heavy transforms baked in; DAP owns · consumers
|   └── reference_data     (DAP loads · consumers SELECT-only)
|       └── <team>         (in-platform team WRITE – their lightweight outputs)

```

Cross-platform consumers (NESO.AI / Palantir) store their lightweight outputs **on their own side**, not in a <team> schema; they may re-ingest into DAP via Pattern B.

Naming conventions (extends the DAP standard):

Object	Pattern	Example
Catalog	<env>_catalog	prod_catalog
DAP governed schemas	acquisition, conformed, reference_data	prod_catalog.conformed
Team schema	<team_snake_case>	prod_catalog.ai_insights

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Team output table	<entity> (curated)	prod_catalog.ai_insights.curated_demand_signals
Sandbox volume	<team>_vol	sandbox_catalog.ai_insights.ai_insights_vol
AD group (writer)	AZ-<Team>-Writer	Az-AllInsights-Writer
AD group (reader)	AZ-<Team>-Reader	az-AllInsights-Reader
Consumer / job SPN	spn-<team>-dap-<env>	spn-aiinsights-dap-prd
SQL Warehouse	<team>-<env>-warehouse	ai-insights-prod-warehouse

5. Onboarding & access provisioning

Two parts: **identity** (AD group via ServiceNow) and **platform** (schema + grants + compute + the consumption-mode access).

5.1 Step-by-step

Step 1 — Consumer raises the Entra ID group request (ServiceNow)

SNOW request to create the Entra ID security groups:

- az-<Team>-Writer — developers who own/write the team schema (in-platform).
- az-<Team>-Reader — people/services that read the team's outputs.
- A **consumer/job SPN** (spn-<team>-dap-<env>) for unattended access (Serve API calls, SQL Warehouse queries, scheduled jobs).

Includes: team, group owner, members, business purpose. Routes to IGA for approval + creation.

Step 2 — Share group + onboarding intake with DAP

Team gives DAP the **AD group names + object IDs (+ SPN object ID)**, plus an intake: data domains/sources needed, **consumption mode(s)** wanted, environments, volumes, cost centre, named data owner, and a **non-PII attestation** for the sandbox. This is the onboarding **Promotion Gate**.

Step 3 — DAP raises a request to Cloud team that syncs identity into Databricks (SCIM)

DAP ensures the AD groups + SPN are **SCIM-provisioned** to the Databricks account, assigned to the workspace, and visible to the metastore. Without this, UC grants cannot resolve.

Step 4 — DAP provisions the platform footprint (IaC / Asset Bundles)

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- (In-platform) creates the <team> schema in every environment + a sandbox UC Volume; sets schema ownership to AZ-<Team>-Writer.
- Creates a **cluster policy** (size caps, auto-terminate, mandatory cost tags) and a **SQL Warehouse (Pro)** – dedicated in Prod.
- Applies **cost tags** (custom_tags:team).

Step 5 – DAP grants access (Unity Catalog + the chosen consumption mode)

- UC grants (own schema + scoped SELECT on Conform/Reference).
- Plus the **consumption-mode access** (SPN reader on the SQL Warehouse, Serve API authorisation rows, and/or a Delta Share).

Step 6 – Consumer starts in the sandbox, develops on sample/BYO non-PII data, and defines the heavy-transform spec.

Step 7 – Data acquisition request (when real data is needed) – DAP onboards the source into Acquisition → curated Conform and extends the SELECT grant / share.

Step 8 – Switchover, consume, light-transform, (optional) re-ingest .

5.2 AD group & SPN design

Identity	Who	What it can do
az-<Team>-Writer	in-platform developers	own + write the team schema (all envs) + sandbox Volume; SELECT on scoped Conform + Reference
az-<Team>-Reader	downstream consumers of the team's outputs	SELECT on the team's <team> schema (prod); optionally scoped Conform read
spn-<team>-dap-<env> (SPN)	apps / jobs / consumption	OAuth client_credentials (no PAT) – calls Serve API, queries the SQL Warehouse, runs jobs

5.3 Identity flow (Entra → Databricks → UC)

ServiceNow request → IGA approves → Entra ID groups + SPN created
 → SCIM provisioning (By Cloud Team) → Databricks Account groups
 → assigned to Workspace + Metastore

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→ Unity Catalog GRANTS + consumption-mode access → access live

5.4 Unity Catalog grants (exact)

Run per environment catalog. UC requires the USE CATALOG → USE SCHEMA → SELECT traversal.

(a) The team's own schema — full ownership/write

```
GRANT USE CATALOG ON CATALOG sndbox_catalog TO `az-AIInsights-Writer`;
ALTER SCHEMA sndbox_catalog.ai_insights OWNER TO `az-AIInsights-Writer`;
GRANT USE SCHEMA, CREATE TABLE, MODIFY, SELECT ON SCHEMA sndbox_catalog.ai_insights TO `az-AIInsights-Writer`;
```

(b) Read access to Conform — scoped to the tables they need (least privilege)

```
GRANT USE CATALOG ON CATALOG prod_catalog TO `az-AIInsights-Writer`;
GRANT USE SCHEMA ON SCHEMA prod_catalog.conformed TO `AZ-AIInsights-Writer`;
GRANT SELECT ON TABLE prod_catalog.conformed.<table_1> TO `AZ-AIInsights-Writer`;
GRANT SELECT ON TABLE prod_catalog.conformed.<table_2> TO `AZ-AIInsights-Writer`;
```

(c) Read access to Reference Data — broad read acceptable (lookups, no PII)

```
GRANT USE SCHEMA ON SCHEMA prod_catalog.reference_data TO `AZ-AIInsights-Writer`;
GRANT SELECT ON SCHEMA prod_catalog.reference_data TO `AZ-AIInsights-Writer`;
```

(d) Sandbox — write to schema + Volume for BYO non-PII data

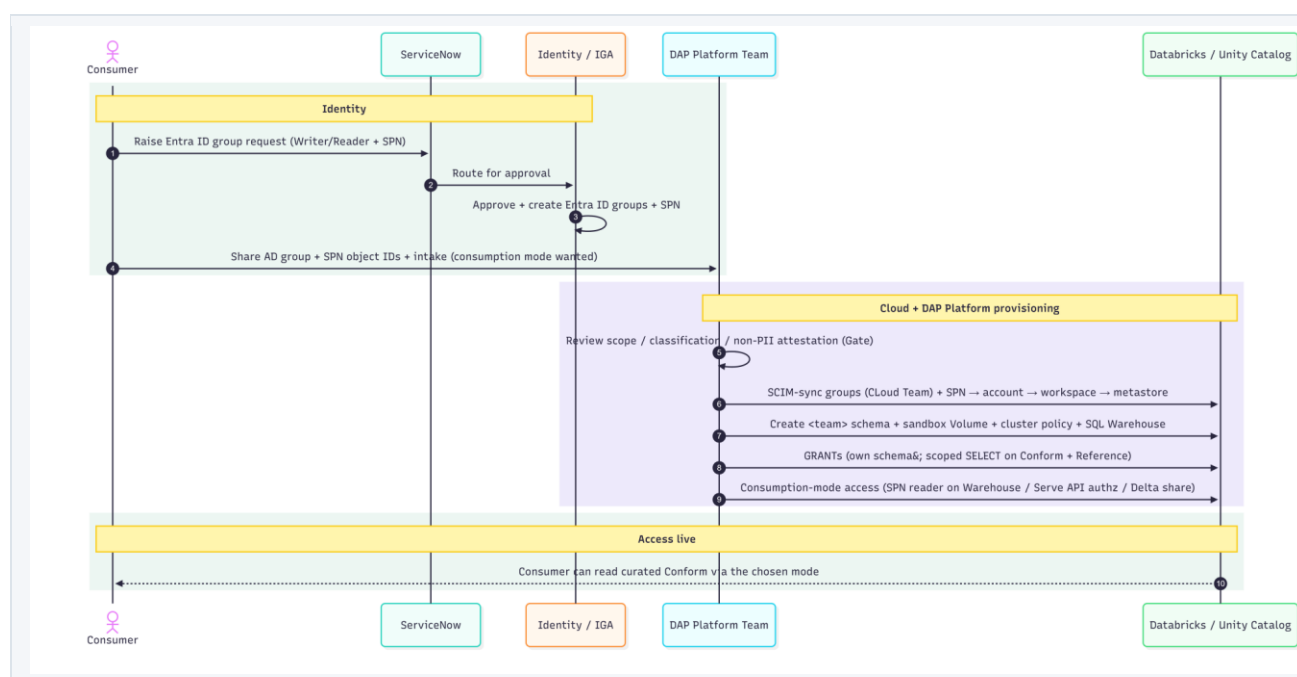
```
GRANT USE CATALOG ON CATALOG sandbox_catalog TO `AZ-AIInsights-Writer`;
ALTER SCHEMA sandbox_catalog.ai_insights OWNER TO `AZ-AIInsights-Writer`;
GRANT USE SCHEMA, CREATE TABLE, MODIFY, SELECT, CREATE VOLUME ON SCHEMA sandbox_catalog.ai_insights TO `AZ-AIInsights-Writer`;
GRANT READ VOLUME, WRITE VOLUME ON VOLUME sandbox_catalog.ai_insights.ai_insights_vol TO `AZ-AIInsights-Writer`;
```

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(e) The consumer/job SPN — grant the SPN (spn-aiinsights-dap-prd) the same scoped SELECT on Conform + Reference as (b)/(c), so apps/jobs/queries run under the SPN, never a user or PAT.

What a consumer does NOT get: any write/MODIFY/CREATE on acquisition or conformed; any access to acquisition (Raw); catalog-wide SELECT on Conform.

5.5 Onboarding sequence (Mermaid)



5.6 Consumption modes (how consumers read DAP data)

Consumers read DAP's curated data through **four governed modes**. Choose by data size + use case. All are SPN-authenticated; none expose raw PII (masks are applied in Conform).

Mode	Best for	Authentication & authorisation	Compute	Status
Serve API	Small to Medium data, programmatic, per-DatasetID	SPN via API ; authorised by the Custom SQL DB per DatasetID (confidential)	DAP SQL Warehouse (configurable)	Live

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		(Paginated Support)		
Direct SQL Warehouse query	big data, AI/ML	consumer SPN with reader access on the SQL Warehouse (granted by DAP) + UC SELECT on Conform	consumer's AI/ML cluster	Live
Delta Sharing	big data, zero-copy, cross-platform	DAP share + consumer recipient	recipient's compute	Design in progress
Power BI	reporting	SPN registered against the SQL Warehouse	Power BI + Warehouse (VNet / On-premise Gateway)	Live

(1) Serve API — small data.

DAP's Serve API (func-dap-consampled-data-prd-uks-01) is fronted by the central **APIM** (apim-dap-consumption-prd-uks-01). The caller authenticates with its **SPN** (OAuth); APIM validates the JWT, rate-limits and IP-filters. Authorisation is **per Dataset ID** via the Custom SQL DB (sql-dap-common-prd-uks-01: dataset_registry, spn_authorisation / user_authorisation) — the **DatasetID** is the primary input in the Serve API URL. The API supports **filter** and **SQL query projection**, and is connected to a **configurable SQL Warehouse**. If a network constraint blocks the URL, the consumer **raises a request to the APIM team** for access. Reference: DAP_Serve_API_HLD.

(2) Direct SQL Warehouse query — big data / AI-ML.

The consumer's AI/ML cluster queries the DAP **SQL Warehouse** directly (databricks-sql-connector / JDBC). The consumer's **SPN must have reader access on the SQL Warehouse** (granted by DAP) plus scoped UC SELECT on the Conform tables. Best for large-volume reads / training-data pulls. Reference: ML_Studio_SQL_Warehouse_.*.

(3) Delta Sharing — big data, zero-copy (design in progress & Strategic for AI/ML and Big Data ex: NESO.AI Instances and Palantir).

The most efficient mode for large data and cross-platform consumers: DAP creates a **share** of the curated Conform tables; the consumer is a **recipient** (Databricks-to-Databricks for in-

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Databricks/Foundry, or token-based for external). No copy, no movement. Reference: Delta_Sharing_SQL_Warehouse_HLD. (In-Progress)

(4) Power BI — reporting.

Reports consume via an **SPN registered against the SQL Warehouse**, through the **VNet Data Gateway**. Reference: PowerBI_VNet_Data_Gateway_HLD.

6. Compute model

Environment	Compute	Notes
Sandbox	Shared policy-bound compute (or serverless)	cost-tagged per team; auto-terminate; quotas
Dev / QA / UAT	Shared policy-bound compute + DAP job clusters	DAP runs the governed pipeline incl. heavy transforms
Prod	Dedicated cluster + dedicated SQL Warehouse (Pro) per team	no Serverless where policy prohibits it ; serves Serve API, direct query, Power BI

- Cross-platform consumers use **their own AI/ML compute** for lightweight transforms.
- Cluster policies cap size, force auto-termination, and **mandate `custom\_tags:team`**.

7. The consumer journey (lifecycle)

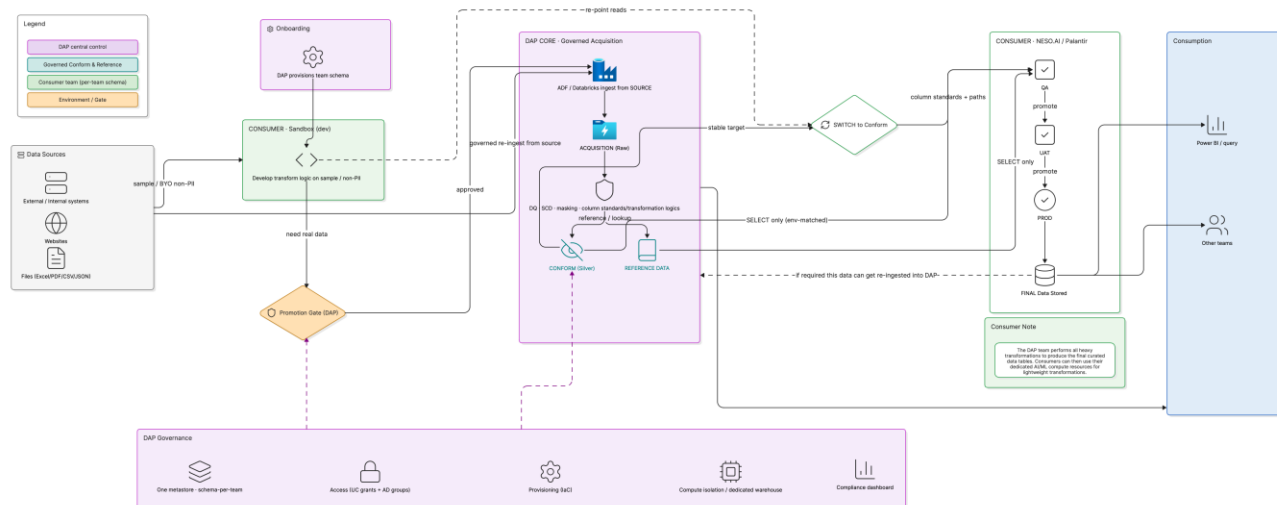
(See the Consumer Journey design flow below.)

- 1. Sandbox dev** — develop **lightweight** transformation logic on sample / BYO non-PII data; define the **heavy-transform spec** for DAP.
- 2. Promotion Gate (DAP)** — request the real source; DAP reviews and acquires it.
- 3. Acquisition → curated Conform (heavy transforms by DAP)** — ADF/Databricks → Acquisition (Raw) → DQ · SCD · masking · column standards · **heavy transformation logic** → curated Conform (Silver) + Reference.
- 4. Switchover** — re-point reads from sandbox tables to **Conform tables**. The only change is adopting DAP's **conformed column standards + table paths** — minimal effort.
- 5. Choose a consumption mode** — Serve API (small) / direct SQL Warehouse (big, AI-ML) / Delta Sharing (big, zero-copy) / Power BI (reporting).

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6. Consume & light-transform — read curated Conform via the chosen mode; apply **lightweight** transforms on own compute; store output (own <team> schema in-platform, or own side cross-platform). In-platform code promotes Dev → QA → UAT → Prod, each env reading the matching Conform env (SELECT-only).

7. (Optional) re-ingest — if the output must become governed / shared, re-ingest into DAP via **Pattern B** (Service Bus → Databricks Delta write) → curated Conform.



8. Data acquisition (how the source gets into curated Conform)

DAP onboards sources through the **canonical Data Acquisition pattern** and applies the **heavy transformation logic** there — never by promoting sandbox data.

Source → ADF / Databricks → Acquisition (Raw) → DQ · SCD1/2 · masking · column standards · heavy transforms → curated Conform (Silver)

Routing rule:

- Energy / transactional data** → conformed (Silver), incremental, SCD + DQ + heavy transforms.
- Reference / lookup data** → reference_data (overwrite, schema-contract enforced, no SCD).

By data type:

Source / type	Pattern	Lands in
Internal/external systems (DB/API), batch	ADF (SHIR if on-prem) → Acquisition → pipeline	conformed

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CSV / JSON payload	ADF or Autoloader → Acquisition	conformed
Excel (lookup)	Reference Data pattern / Generic Excel Extractor + JSON config	reference_data
PDF / unstructured	OCR → text/embeddings → Delta	conformed
Websites	ADF REST/web connector or approved scraper — legal/compliance sign-off	conformed

PII is masked in Conform via UC Column Masks; consumers inherit the mask and never see raw PII.

9. Promotion / CI-CD

For **in-platform** consumers, the **lightweight transformation code** is the unit that moves through environments; **curated Conform is the stable target** at each tier.

```
Git repo (team-owned) → Databricks Asset Bundles
→ deploy Dev → QA → UAT → Prod (DAP CI/CD SPN promotes prod)
→ each env reads matching Conform env (prod_catalog.conformed in Prod, etc.)
→ writes output into <env>_catalog.<team>
```

Cross-platform consumers promote within **their own** platform's lifecycle and read the matching DAP Conform env via their consumption mode.

10. Security & network

- **Zero public internet** — all platform access via Private Endpoints + Private DNS.
- **SPN OAuth `client\_credentials`** for all service identities — **no PAT tokens**.
- **APIM** in front of the Serve API — validate-jwt, rate-limit, ip-filter, cors, log-to-eventhub.
- **Sandbox is non-PII**; PII only via the governed pipeline with UC Column Masks in Conform.
- **Least privilege** — scoped SELECT on Conform; SPN reader (not admin) on the SQL Warehouse; no access to Acquisition (Raw).

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- **Audit everywhere** — `system.access.audit` (UC) + Serve API `access_audit_log` (Custom SQL DB).

11. Governance, observability & compliance

- **Access** — UC grants + AD groups + Serve API DatasetID authorisation; ownership of governed schemas stays with DAP.
- **Provisioning** — everything as code (Terraform / Asset Bundles).
- **Lineage** — `system.access.table_lineage` / `column_lineage` proves consumer outputs derive from governed Conform (SOR compliance check).
- **Cost / chargeback** — `custom_tags:team` maps compute + DBU per team; 120%-of-budget flag.
- **Compliance dashboard** — `DAP_Governance_Dashboard_Queries.sql`, schema = team boundary.

12. Guardrails (non-negotiable)

1. **No write on Acquisition or Conform** for any consumer — SELECT-only / share-only, ever.
2. **Scoped grants** — SELECT on specific Conform tables, not catalog-wide.
3. **No access to Acquisition (Raw)** — consumers start at curated Conform.
4. **DAP owns heavy transforms** — consumers do lightweight only; heavy logic is productionised into the DAP pipeline, not run ad-hoc by consumers.
5. **Re-ingest from source** — never promote sandbox data to Conform (reference data via the loader is the sole exception).
6. **Sandbox is non-PII only**; sandbox reads QA/UAT only, never Prod.
7. **No PAT tokens** — SPN OAuth for all service/job/consumption identities.
8. **Serve API DatasetID is confidential** — authorised per SPN in the Custom SQL DB.

13. Roles & responsibilities (RACI)

Activity	DAP (Central)	Consumer	Cloud
Raise Entra ID group + SPN (SNOW)	C	R	I
Approve & create AD group / SPN (IGA)	C	I	R

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Create schema, grants, compute, Volume, consumption-mode access	R	I	I
SCIM / workspace entitlement	C	I	R
Onboard source → Acquisition → curated Conform (heavy transforms)	R	C	NA
Define data model / DQ rules / heavy-transform spec	C	R	NA
Build & run heavy / big transformations → curated tables	R	C	
Lightweight transforms on own compute + final output	I	R	
(Optional) re-ingest output into DAP (Pattern B)	R	C	
Access reviews, compliance, cost monitoring	R	C	

*(R = Responsible, C = Consulted, I = Informed) *

14. Onboarding checklist

Consumer brings:

- ☐ SNOW request for AZ-<Team>-Writer / -Reader + consumer/job SPN

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- ❑ Intake: domains/sources, **consumption mode(s)** wanted, envs, volumes, cost centre, data owner
- ❑ Non-PII attestation for sandbox
- ❑ Data model + DQ rules + **heavy-transform spec** (at the data Promotion Gate)
- ❑ (In-platform) Git repo for their lightweight transformation code

DAP provisions:

- ❑ SCIM sync (Cloud Team) + workspace/metastore assignment of AD groups + SPN
- ❑ <team> schema (in-platform) + sandbox Volume
- ❑ Schema ownership → Writer; scoped SELECT on Conform + Reference
- ❑ Cluster policy + dedicated Prod SQL Warehouse (Pro) + cost tags
- ❑ **Consumption-mode access**: SPN reader on the Warehouse / Serve API authorisation rows (DatasetID) / Delta share
- ❑ Source onboarding into Acquisition → curated Conform (heavy transforms)
- ❑ Registry entry + governance dashboard coverage

15. Risks & mitigations

ID	Risk	Mitigation
R1	Consumer requests catalog-wide Conform access	Scoped, per-table SELECT; ARB review
R2	PII leaks into sandbox	Non-PII attestation + scanning; PII only via governed pipeline + masking
R3	Sandbox data promoted to prod by shortcut	Re-ingest from source; lineage check blocks sandbox→prod dependency
R4	Consumer runs heavy transforms ad-hoc, fragmenting logic	Heavy transforms productionised into DAP pipeline; consumers light only
R5	Noisy-neighbour compute contention	Per-team cluster policies + dedicated Prod warehouse + cost caps

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R6	Conform schema/column change breaks consumer code	Conform published with a contract + change notice; stable column standards
R7	Serve API DatasetID leaks	DatasetID confidential; per-SPN authorisation in Custom SQL DB; APIM JWT + rate-limit
R8	Orphaned access after team/member changes	Periodic UC access reviews; AD-group-driven (remove from group = remove access)

Appendix A — SNOW request template (AD group + SPN)

```
Request type:      Entra ID security group + SPN creation
Groups:           AZ-<Team>-Writer, AZ-<Team>-Reader
SPN:              spn-<team>-dap-prd (OAuth client_credentials)
Group owner:      <team lead UPN>
Initial members:  <UPNs>
Purpose:          DAP platform access for <Team>
Consumption mode: Serve API | Direct SQL Warehouse | Delta Sharing | Power BI
Data domains:     <e.g. demand, network, ...>
Environments:     sandbox, dev, qa, uat, prod
Cost centre:      <code>
Data owner:       <name>
Non-PII attest.:  Yes — sandbox will hold non-PII data only
```

Appendix B — Reference diagrams & HLDs

- **Serve API** — [HLD](#)
- **Delta Sharing** — [Delta_Sharing_SQL_Warehouse_HLD.docx](#) (In-Progress)
- **ML Studio** → **SQL Warehouse** — [ML_Studio_SQL_Warehouse_*](#)
[Databricks workspace connectivity from ML workspace - Overview](#)

Appendix C — Glossary

- **Acquisition** — Raw / Bronze landing layer (DAP-owned, SPN-only).
- **Conform / Conformed** — curated Silver layer: cleansed, conformed, DQ-enforced, masked, **with heavy transformation logic baked in** (DAP-owned; consumers SELECT-only / share-only).
- **Reference Data** — governed lookup tables (overwrite load, no SCD).

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- **Consumption mode** — a governed channel to read DAP data: Serve API, direct SQL Warehouse, Delta Sharing, Power BI.
- **Dataset ID** — confidential identifier keying Serve API authorisation in the Custom SQL DB.
- **In-platform consumer** — works inside DAP (schema + dedicated compute). **Cross-platform consumer** — works on its own platform (e.g. NESO.AI / Palantir Foundry), reads DAP via a consumption mode.
- **Promotion Gate** — DAP review point: (onboarding) for access, (data) before acquiring a source.
- **Switchover** — the consumer re-pointing reads from sandbox to Conform tables.
- **Pattern B** — the re-ingest feedback path (consumer output → Service Bus → Delta write → Conform).
- **Umbrella** — the single shared Unity Catalog metastore.